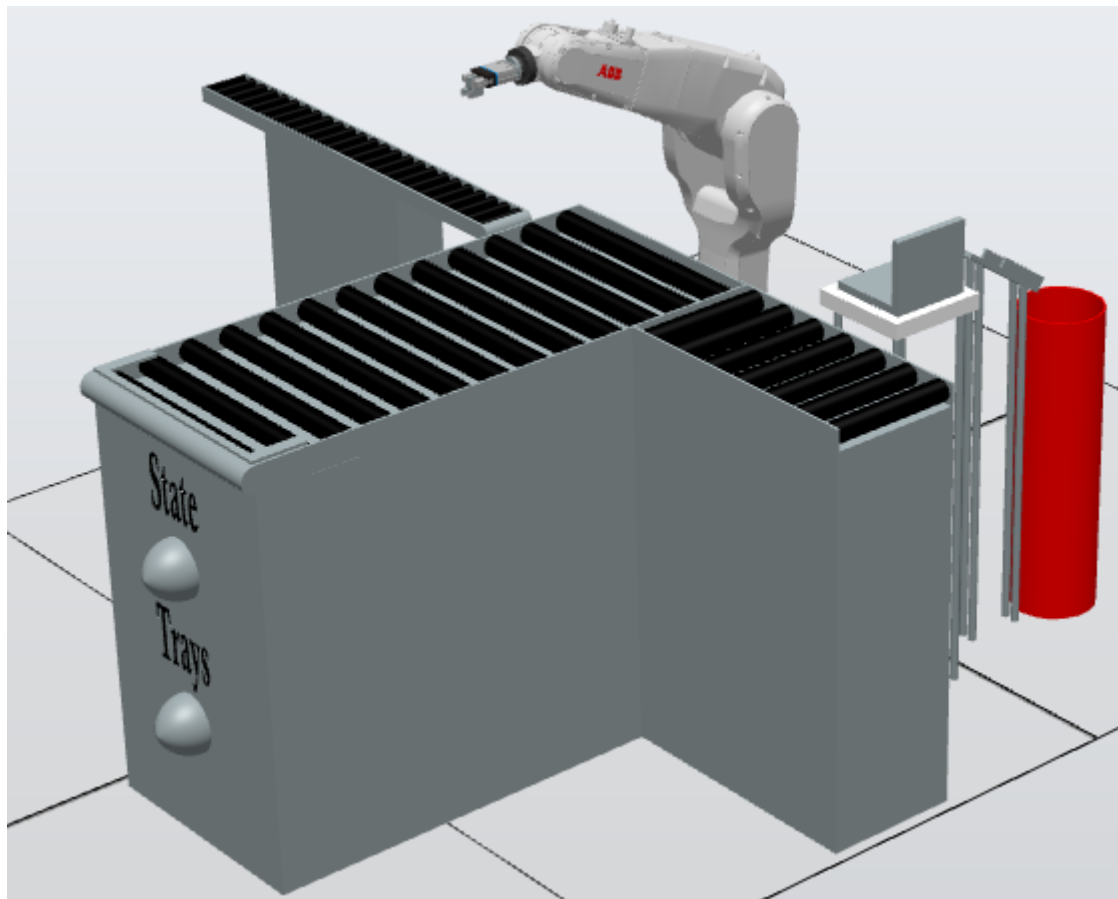


Automated Pick & Place Cell



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1. Executive Summary

This project presents the development of an automated pick-and-place cell for the handling and classification of automotive workpieces. The system was developed as an independent engineering project and implemented entirely in a simulation environment, combining mechanical design, industrial automation and robotic programming.

The cell was designed around an ABB IRB 1200 robot and an automotive bracket as the workpiece. The process includes workpiece feeding, robotic picking, a simulated inspection stage and subsequent classification into OK or NOK parts. The simulated inspection assigns approximately 90% of the workpieces to the OK category and 10% to the NOK category. OK parts are placed into trays with six positions, while NOK parts are transferred to an inclined platform from which they can move into a dedicated container. A five-tray buffer system was also developed to enable semi-automatic tray replacement and continuous operation.

The control system was developed using Siemens TIA Portal V20 and PLCSIM Advanced, with the PLC logic programmed in Ladder Diagram (LAD). WinCC was used to develop the operator interface. The robotic cell was implemented in ABB RobotStudio using RAPID, while OPC UA was used to establish bidirectional communication between the simulated PLC and the robot environment. This allowed process signals, counters and operating states to be exchanged between both systems during operation.

Several engineering challenges were addressed during development, including robot-tray collision problems, automatic tray management, communication between two computers and timing synchronisation between the PLC and robot. The complete simulated cell sequence, including normal production, tray replacement, OK/NOK handling, End-of-Cycle, Quick Stop and Emergency Stop behaviour, was tested at different stages of the process.

The resulting system demonstrates the integration of mechanical design, PLC programming, HMI development, robotic programming and industrial communication within a single automated manufacturing concept. However, the project remains limited to simulation: no physical hardware, real inspection system or industrial safety system was implemented or validated. These aspects, together with manufacturing-ready mechanical design and further improvements to the system, are identified as future development opportunities.

2. Project Objectives and Scope

2.1 Main Objective

The main objective of the project was to develop an automated pick-and-place manufacturing cell in a simulation environment. The system was designed to combine robotic handling, PLC-based process control, simulated workpiece classification and semi-automatic tray management within a single cell.

The project also aimed to establish communication between the robotic and PLC environments so that both systems could operate as part of the same process. The final system integrates mechanical design, robot programming, PLC programming, HMI development and OPC UA communication.

2.2 Specific Objectives

The following objectives were defined for the development of the cell:

- Design the main components and layout of the cell using Siemens NX.

- Develop the PLC control logic in Siemens TIA Portal using Ladder Diagram (LAD).
- Design the process logic using GRAFCET before implementing it in the PLC program.
- Develop the operator interface using WinCC.
- Program the ABB robot and the required process movements in RobotStudio using RAPID.
- Establish bidirectional communication between the simulated PLC and RobotStudio using OPC UA.
- Implement the simulated classification of workpieces into OK and NOK categories.
- Develop the OK-piece tray filling system, with six positions per tray.
- Implement the five-tray buffer and semi-automatic tray replacement process.
- Implement End-of-Cycle, Quick Stop and Emergency Stop operating behaviour.
- Test the implemented functions and the complete cell sequence within the simulation environment.

2.3 Project Scope and Limitations

The project was intentionally limited to a simulation-based implementation. No physical robot, PLC, conveyor, sensor, gripper or other hardware was manufactured or commissioned.

The inspection stage was also simulated. The 90% OK and 10% NOK distribution represents a programmed classification probability and does not correspond to the accuracy of a real inspection system. No camera-based vision system was implemented, and no artificial intelligence or machine learning method was included in the final system.

The mechanical design was developed for simulation, spatial integration and robot operation. It was not intended as a manufacturing-ready design and therefore does not include complete manufacturing drawings, detailed tolerances, material specifications or structural calculations.

Finally, the emergency handling implemented in the simulation does not constitute a validated industrial safety system. Formal safety implementation and validation were outside the scope of the project.

3. System Overview

3.1 Cell Layout and Main Components

The developed system represents an automated manufacturing cell for the handling and classification of an automotive bracket. The complete cell was implemented in a simulation environment and consists of a workpiece input conveyor, an ABB robot, a simulated scanning station, an OK tray conveyor, an NOK platform and a five-tray buffer system.

The main robotic system is an ABB IRB 1200_5_90_STD_03, implemented in RobotStudio 2026.2 with RobotWare 6.07.01. The robot is responsible for handling the workpiece throughout the main process. The mechanical components surrounding the robot were designed in Siemens NX 2212.

The cell receives the workpiece from another conveyor. The input conveyor transports the bracket to the robot working area, where a pick sensor detects its position and stops the conveyor. The robot can then pick the workpiece and start the handling sequence.

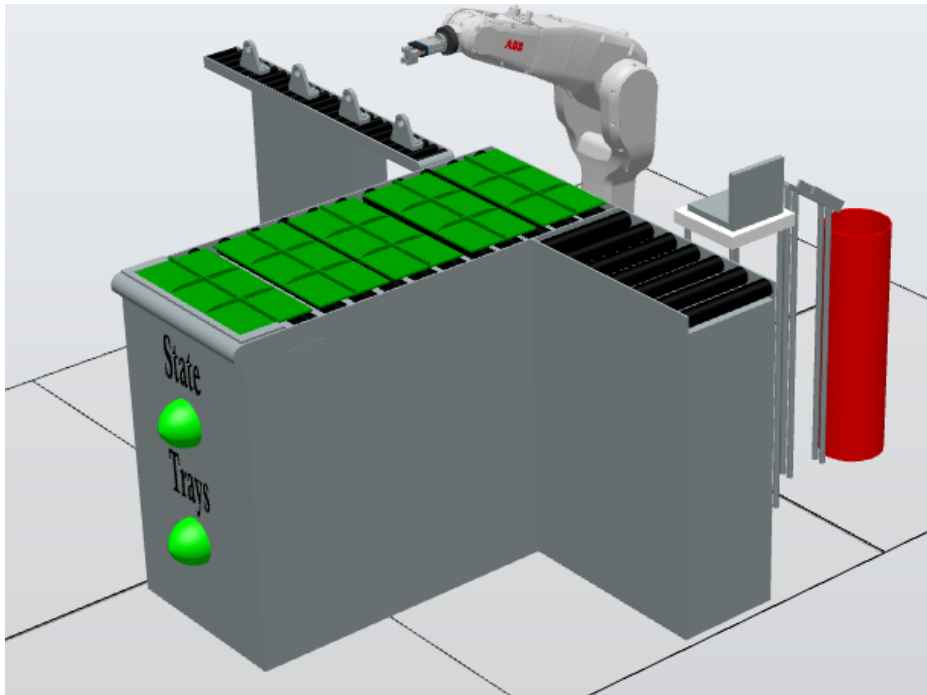


Figure 1. Cell Layout.

3.2 Workpiece Handling and Inspection

Once the workpiece is detected, the robot picks it using the custom-designed gripper and moves it to the scanning station. The inspection process is simulated rather than performed using a physical sensor or vision system.

The robot places the workpiece in the simulated scanner and waits for three seconds. After this period, the classification result is generated in the RAPID program. A random value is generated to simulate the inspection result, with approximately 90% of the workpieces classified as OK and 10% as NOK.

The robot then picks the workpiece from the scanner again and moves it to the corresponding destination according to the classification result.

3.3 OK and NOK Handling

When a workpiece is classified as OK, the robot places it in the active tray. Each tray contains six positions arranged in two rows and three columns. The position used by the robot is determined by the current tray position counter.

When a workpiece is classified as NOK, the robot places it on a dedicated inclined platform. The platform has an inclination of 35°, allowing the workpiece to move into the NOK container by gravity. Therefore, the robot does not need to place the NOK workpiece directly inside the container.

3.4 Semi-Automatic Tray Buffer

The semi-automatic tray buffer system consists of five trays arranged in a queue. During the initial conditions, the operator places the five trays into the buffer one by one. Sensors along the tray conveyor detect their positions and control their movement.

The tray located at the filling position receives the OK workpieces from the robot. Once all six positions are filled, the tray exit conveyor is activated and the full tray leaves the filling position. The next tray in the queue is then moved into the filling position.

The OK tray conveyor has an L-shaped configuration. Its vertical section is used as the tray buffer, while its horizontal section is used to move the full trays out of the cell. The system therefore allows the five trays to be used sequentially without requiring the operator to replace each tray individually.

After all five trays have left the cell, the operator must repeat the tray filling process before production can continue.

3.5 Overall Process Flow

The complete process can be summarised as follows:

1. The operator starts the system and loads the five empty trays into the buffer.
2. The input conveyor transports an automotive bracket into the robot working area.
3. The pick sensor detects the workpiece and the conveyor stops.
4. The robot picks the workpiece and places it in the simulated scanner.
5. The robot waits three seconds while the simulated inspection is completed.
6. The workpiece is classified as OK or NOK.
7. The robot picks the workpiece again and places it in the corresponding location.
8. If the workpiece is OK, it is placed in the current tray position. If it is NOK, it is placed on the inclined NOK platform.
9. When all six positions of the current tray are filled, the tray is moved out of the filling position.
10. The next tray in the buffer is moved into the filling position.
11. The process continues until all five trays have left the cell.

4. Control System Architecture

4.1 Control Architecture

The control architecture was developed using two separate computer environments. The first computer contains the Siemens automation environment, while the second computer contains the ABB robotic environment. Both systems are connected through the same router and exchange process information during the simulation.

The Siemens environment consists of TIA Portal V20, a simulated S7-1500 CPU 1511-1 PN running in PLCSIM Advanced V7.0, WinCC for the HMI and UaExpert for monitoring the OPC UA communication.

The robotic environment consists of ABB RobotStudio 2026.2 with an IRB 1200_5_90_STD_03 robot and RobotWare 6.07.01. The robot sequence is programmed in RAPID.

The overall architecture is therefore divided into two main control environments:

- **Siemens environment:** process control, sequence management, counters, timers and HMI operation.
- **ABB RobotStudio environment:** robot movement, workpiece handling and robotic process sequence.

OPC UA provides the communication link between both environments.

4.2 PLC and RobotStudio Interaction

The PLC and the robot operate together during the complete process. The PLC manages the general state of the cell and provides the signals required to coordinate the different process steps. RobotStudio executes the corresponding robot movements and returns information about the progress of these operations.

For example, when a workpiece reaches the pick position, the PLC detects the corresponding signal and the robot can execute the picking operation. After the robot completes a specific operation, signals are returned to the PLC so that the next process step can be activated.

This interaction is also used for tray management, workpiece counters and process states. The PLC therefore acts as the main process-control element, while RobotStudio performs the robotic operations required by the sequence.

4.3 OPC UA Communication

Bidirectional OPC UA communication was implemented to exchange information between the simulated PLC and RobotStudio.

The simulated S7-1500 PLC acts as the OPC UA server. RobotStudio acts as the OPC UA client through the OpcuaClient Station Logic component. UaExpert was used to monitor and verify the exchanged OPC UA variables during the development and testing of the system.

The communication includes signals related to workpiece handling, tray positions, classification and process states. Information generated in RobotStudio is transferred to the PLC, while PLC variables are also made available to the robotic environment.

The final configuration uses Ethernet for the computer running the OPC UA server and Wi-Fi for the RobotStudio computer, with both computers connected to the same router. This configuration allowed the OPC UA communication to operate correctly between the two environments.

5. PLC and HMI Development

5.1 PLC Programming and GRAFCET

The PLC control system was developed in Siemens TIA Portal V20 using Ladder Diagram (LAD). Before implementing the program, the main process logic was designed using GRAFCET. This approach was used to define the sequence of operations and the transitions between the different operating states before transferring the logic to the PLC program.

GRAFCET was mainly used for the conveyor control and the state-light logic. The robot and gripper movements were not implemented using GRAFCET, as these operations are controlled directly in RobotStudio through the RAPID program. The tray light logic was also implemented separately because it is based on the tray counter. Quick Stop behaviour was handled directly in the PLC program rather than through the GRAFCET structure.

The PLC program contains four main GRAFCET sequences:

- **GSequence:** main sequence that manages the different operating modes.
- **GInitCond:** controls the initial conditions and the tray-filling process.
- **GNormProd:** controls the automatic production sequence.
- **GEmergency:** manages the emergency-stop sequence.

GSequence activates the corresponding sequence depending on the current operating condition of the cell. After the initial conditions have been completed, the system can enter Normal Production. End-of-Cycle is also managed through the main sequence.

5.2 PLC Software Structure

The PLC software was structured into Function Blocks (FBs), Functions (FCs) and their associated data blocks. This modular structure separates the different control tasks and makes the program easier to analyse and modify.

The main program structure called from OB1 is:

- **FB_Sequence + DB_FB_Sequence:** management of the main process states.
- **FB_InitCond + DB_FB_InitCond:** tray-filling sequence.
- **FB_NormProd + DB_FB_NormProd:** automatic production sequence.
- **FB_Emergency + DB_FB_Emergency:** emergency-stop sequence.
- **FC_VarControl:** synchronisation variables between the different function blocks.
- **FC_Outputs:** process outputs.
- **FC_Count:** process counters.
- **FC_Temp:** process timers.
- **FC_Communication:** preparation of variables for OPC UA communication.

OB100 is used during PLC start-up to initialise the End-of-Cycle and Emergency signals to 1. These signals correspond to normally closed buttons and therefore require this initialisation to represent their normal operating condition.

5.3 Functional Blocks and Functions

FB_Sequence contains the steps associated with the different operating modes of the system, including Initial Conditions, Normal Production and End-of-Cycle.

FB_InitCond controls the tray-filling process. It manages the movement of the trays through the buffer until the required initial configuration is reached.

FB_NormProd contains the logic for the automatic production mode. It coordinates the workpiece conveyor, workpiece classification and tray replacement.

FB_Emergency contains the sequence used to handle the emergency-stop condition and the subsequent rearming of the system.

FC_VarControl contains the internal variables used to synchronise the different function blocks. For example, it manages the signals indicating the current operating mode, completion of the initial conditions, tray exit and emergency status.

FC_Outputs centralises the main process outputs. These include the tray conveyor movement, workpiece conveyor movement, robot Pick and Scan commands, OK/NOK destination commands and the emergency alarm.

FC_Count manages the counters used during the process. CTU counters are used for the number of processed OK and NOK workpieces. CTUD counters are used for the current position within the tray and the number of trays remaining in the buffer.

FC_Temp contains the six TON timers used by the PLC program. These timers provide the required delays for different process transitions and help synchronise the PLC logic with the simulated robotic sequence.

FC_Communication prepares PLC variables for the OPC UA exchange. MOVE instructions are used to assign the required values to the Light_State and Light_Tray variables. The current values of the counters are also transferred to memory-word variables so that they can be exchanged through OPC UA.

5.4 PLC Signals and Process Synchronisation

Communication signals between RobotStudio and the PLC are used to synchronise the two simulation environments. RobotStudio provides information about workpiece detection, tray positions and the completion of robotic operations, while the PLC provides the corresponding process commands and status information.

The exchanged signals can be divided into two main groups: signals sent from RobotStudio to the PLC and signals sent from the PLC to RobotStudio. In addition, the PLC uses internal variables and HMI commands to manage the overall process state.

5.4.1 RobotStudio to PLC Signals

RobotStudio sends signals to the PLC to indicate the status of the workpiece, trays and robotic operations. These signals are used by the PLC to determine whether the corresponding process step has been completed and whether the next operation can be activated.

The main signals exchanged from RobotStudio to the PLC are:

Signal	Type	Function
Piece_OK	Bool	Indicates whether the classified workpiece is OK or NOK.
Pick_Sensor	Bool	Indicates that a workpiece has reached the robot pick position.
Tray_Sensorinit	Bool	Indicates that a tray has been detected at the initial tray position.
Tray_Sensor1	Bool	Indicates that a tray has reached the first buffer position.

Tray_Sensor2	Bool	Indicates that a tray has reached the second buffer position.
Tray_Sensor3	Bool	Indicates that a tray has reached the third buffer position.
Tray_Sensorend	Bool	Indicates that a tray has reached the final filling position.
Piece_Tray_Done	Bool	Indicates that the robot has completed the OK-piece placement path.
Piece_Bin_Done	Bool	Indicates that the robot has completed the NOK-piece placement path.
Pick_Done	Bool	Indicates that the robot has completed the picking operation.
Tray_Out	Bool	Indicates that the full tray has left the cell.
Scan_Done	Bool	Indicates that the robot has completed the scanning sequence.

Table 1. RobotStudio-PLC exchanged signals

5.4.2 PLC to RobotStudio Signals

The PLC also sends process commands and status information to RobotStudio. These signals allow the robotic sequence to respond to the current operating state of the PLC and execute the required operation.

The main signals exchanged from the PLC to RobotStudio are:

Signal	Type	Purpose
Run	Bool	Indicates the current state of the Run input.
EoC	Bool	Indicates the current state of the End-of-Cycle input.
EM	Bool	Indicates the current state of the Emergency condition input.
Rearm	Bool	Indicates that the emergency condition has been rearmed.
Stop	Bool	Indicates that the Stop command is active.
Light_State	Int	Defines the colour displayed by the State lights.
Light_Tray	Int	Defines the colour displayed by the tray status lights.
Tray_Hole	Int	Indicates the tray position where the next OK workpiece must be placed.
Move_Tray_Conveyor	Bool	Commands the tray-filling section of the conveyor to move.
Move_Piece_Conveyor	Bool	Commands the workpiece input conveyor to move.
Move_Tray_Conveyor_Exit	Bool	Commands the tray exit section of the conveyor to move.

Signal	Type	Purpose
OK_Piece	Bool	Commands RobotStudio to execute the OK-piece handling path.
NOK_Piece	Bool	Commands RobotStudio to execute the NOK-piece handling path.
Pick	Bool	Commands RobotStudio to execute the pick operation.
Scan	Bool	Commands RobotStudio to execute the scanning operation.
Tray_Left	Int	Indicates the number of trays remaining in the buffer.
Piece_Tray	Int	Indicates the number of OK workpieces currently placed in the active tray.

Table 2. PLC-RobotStudio exchanged signals

5.5 Counters and Timers

Several counters are used to manage the production process and tray system.

The contPiece counter records the number of OK pieces placed in the current tray and is reset when the tray changes. The contOK and contNOK counters record the total number of OK and NOK workpieces processed during the operation.

The contHole counter identifies the current position within the active tray. It is used by the robotic sequence to determine where the next OK workpiece has to be placed. The counter therefore links the PLC process state with the corresponding tray position in RobotStudio.

The contTray counter represents the number of trays remaining in the buffer. Its value is also used to determine the tray-status indicator shown on the HMI.

Six TON timers are used in the PLC program. Their preset times include 1 s and 50 ms values, depending on the required process transition. These timers were mainly used to provide controlled delays and avoid timing conflicts between the PLC sequence and RobotStudio.

5.6 HMI Development

The operator interface was developed using WinCC, integrated into TIA Portal V20. The simulated HMI is based on a TP900 Comfort panel.

The HMI was designed to provide the operator with the controls required to operate the simulated cell and to display its current status. Operator interaction is intentionally limited to five buttons:

- **Run:** starts the system.
- **Stop:** requests a Quick Stop.
- **EM:** requests an Emergency Stop.
- **EoC:** requests an End-of-Cycle stop.
- **Rearm:** rearms the system after an emergency condition.

The HMI also provides visual feedback for these controls. The Run, EM, EoC, Stop and Rearm indicators change between active and inactive states according to the corresponding PLC signals.

The main process-status indicators are the State Light and Tray Left Light. The State Light represents the current state of the cell, while the Tray Left Light indicates the number of trays remaining in the buffer. The latter is generated from the contTray counter.

The HMI also displays the following process values:

- **OK Pieces:** current value of contOK.CV.
- **NOK Pieces:** current value of contNOK.CV.
- **Tray Left:** current value of Tray_Left.
- **Pieces in Tray:** current value of Piece_Tray.
- **Current Hole:** current value of Tray_Hole.

The operator cannot modify these displayed process variables directly. Interaction with the system is restricted to the five control buttons, while the remaining elements provide status and process information.¹

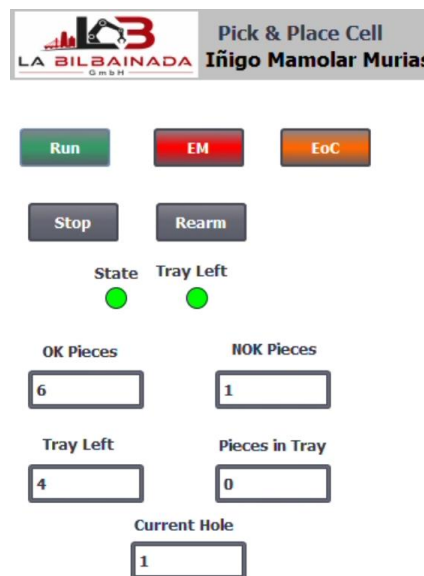


Figure 2. HMI Display

6. RobotStudio Programming

6.1 RobotStudio and Robot Configuration

The robotic part of the cell was developed in ABB RobotStudio 2026.2 using an ABB IRB 1200_5_90_STD_03 robot and RobotWare 6.07.01. The robot was programmed using the RAPID programming language.

RobotStudio was used to simulate the robot movements and the interaction between the robot, the workpiece, the gripper and the surrounding cell components. The robotic program was also

¹ Note: “LA BILBAINADA GmbH” is a fictitious company name used for presentation purposes

connected to the PLC through the OPC UA communication described in Section 4, allowing the robot sequence to be coordinated with the PLC-controlled process.

6.2 RAPID Program Structure

The RAPID program is organised into process procedures and interrupt routines. The main program coordinates the execution of the different robotic operations, while the individual procedures contain the movements and signal operations associated with each stage.

The main program executes the following procedures during the normal production sequence:

1. SIGNALINIT
2. CONDINIT
3. PIECECONVEYOR
4. PICKPIECE
5. SCANPIECE
6. OKNOK

The SIGNALINIT procedure is executed first and establishes the initial state of the robotic system. It resets the main process feedback signals, including Piece_Bin_Done, Piece_Tray_Done, Scan_Done, Piece_OK and Pick_Done. It also establishes the initial state of the gripper and the simulated status lights. In addition, the procedure connects the End-of-Cycle, Emergency Stop and Quick Stop inputs to their corresponding interrupt routines.

After the initialisation, the CONDINIT procedure manages the initial tray-filling stage. It controls the positioning of the trays until the required initial configuration of the buffer has been established. The TRAYFILL procedure is used within this stage to introduce and position the trays sequentially. The tray sensors provide the position information required to determine when the conveyor should be stopped or restarted.

Once the initial conditions have been completed, the remaining process procedures are executed cyclically during normal operation. PIECECONVEYOR, PICKPIECE, SCANPIECE and OKNOK correspond to the main stages of the robotic production sequence. Their detailed operation is described in Section 6.3.

In addition to the main process procedures, dedicated procedures are used for the two classification paths. OKPIECE manages the handling of OK workpieces, while NOKPIECE manages the NOK workpiece path.

The program also contains three interrupt routines:

- EoCycle, for End-of-Cycle handling.
- EmerStop, for Emergency Stop handling.
- tSTOP, for Quick Stop handling.

The interrupt routines are connected to their corresponding inputs during SIGNALINIT. Their operation is described in Section 6.5.

This structure separates the initialisation, production sequence and stop-condition handling into independent functional elements while maintaining a single overall robotic program.

6.3 Robot Pick-and-Place Sequence

During normal production, the robot performs the workpiece handling sequence through a series of dedicated procedures.

The PIECECONVEYOR procedure manages the arrival of the workpiece to the robot working area. The workpiece is transported until the pick position is reached, allowing the robot to start the handling sequence.

The PICKPIECE procedure performs the first robotic handling operation. The robot moves through the required approach and picking positions, activates the gripper and verifies the corresponding gripper condition. Once the operation is completed, the Pick_Done signal is provided to the PLC.

The workpiece is then transferred to the simulated scanning station through the SCANPIECE procedure. The robot places the workpiece at the scanner position and waits for the defined inspection period. In the simulation, this period is three seconds. The classification result is subsequently generated in the RAPID program and communicated through the Piece_OK signal.

The OKNOK procedure determines which handling path is executed according to the classification result. For an OK workpiece, the OKPIECE procedure moves the robot to the active tray and places the workpiece in the required tray position. After an OK placement, the Piece_Tray_Done signal is sent to the PLC. For an NOK workpiece, the NOKPIECE procedure moves the robot to the inclined NOK platform and releases the workpiece onto it, sending the Piece_Bin_Done signal to the PLC.

The individual procedures therefore correspond to the main robotic operations of the cell, while the PLC signals are used to indicate the completion of the relevant operations and coordinate the following process step.

6.4 Tray Handling and Robot Coordination

Tray handling requires coordination between the PLC-controlled tray system and the robot positioning sequence. Each OK tray contains six positions arranged in two rows and three columns. The position required for the next OK workpiece is communicated to RobotStudio through the Tray_Hole variable.

The PLC maintains the current tray position information and provides the corresponding Tray_Hole value to RobotStudio through OPC UA. The robot uses this information to select the appropriate target for the OK-piece placement. Therefore, the tray position is determined by the PLC rather than independently by the robot.

After completing the placement, RobotStudio sends the Piece_Tray_Done signal to the PLC. The PLC can then update the corresponding counters and determine the position required for the next OK workpiece. This creates a feedback sequence in which the PLC provides the target position, RobotStudio performs the placement and RobotStudio returns the completion signal.

When the sixth OK workpiece has been placed, the PLC detects that the active tray is full and initiates the tray replacement sequence. The full tray is moved towards the exit section of the tray conveyor, and the next tray in the buffer is subsequently positioned in the filling area.

The tray buffer is controlled using the tray position sensors. These sensors provide the PLC with the position of the trays and allow the conveyor movements to be activated or stopped at the required positions.

The coordination between the tray position data, robot targets and PLC signals allows the robotic placement sequence and the automatic tray replacement process to operate as a single coordinated system.

6.5 Interrupt Routines and Stop Conditions

Three interrupt routines were implemented in the RAPID program to manage End-of-Cycle, Emergency Stop and Quick Stop conditions. The routines are connected to the corresponding digital inputs during the initial signal configuration in SIGNALINIT.

6.5.1 End-of-Cycle

The EoCycle interrupt is used to manage the End-of-Cycle request during normal production. When the corresponding input is activated, the interrupt changes the robotic status from normal operation to the End-of-Cycle condition by changing the associated status light. The main production loop can then continue until the current cycle has been completed before the cell stops.

This behaviour allows the operator to request the end of production without immediately interrupting the workpiece currently being processed.

6.5.2 Emergency Stop

The EmerStop interrupt handles the Emergency Stop condition. When the emergency input is activated, the normal status indication is disabled and the emergency indication is activated. The emergency light is flashed while the system remains in the emergency condition.

The routine waits until the Rearm signal is activated. After rearming, the robot waits for the Run command before continuing operation. The status light is then restored according to the current process condition, including the End-of-Cycle condition if it was already active.

The implemented behaviour allows the simulated process to resume from the point at which the emergency condition occurred. This functionality was tested at different points of the process within the simulation environment.

It should be noted that this implementation represents emergency handling within the simulated cell and does not constitute a validated industrial safety function.

6.5.3 Quick Stop

The tSTOP interrupt is used for the Quick Stop function. When the corresponding stop input is activated, the robotic sequence is interrupted and remains stopped until the Run command is activated again.

Unlike the Emergency Stop routine, no emergency rearming sequence is required. The purpose of the Quick Stop is therefore to temporarily interrupt the process while allowing it to continue from the same point after the Run command is received.

The three interrupt routines provide different stopping behaviours according to the operating condition: End-of-Cycle allows the current production cycle to finish, Quick Stop temporarily pauses the process, and Emergency Stop activates the simulated emergency handling sequence.

7. Process Sequence and Operating Modes

7.1 Initial Conditions

Before normal production can start, the cell must first reach the required initial conditions. The tray buffer is filled with five empty trays, which are introduced sequentially using the tray conveyor and the corresponding tray position sensors.

The tray filling sequence continues until a tray reaches the final filling position. The trays are then arranged in the buffer so that they can be supplied sequentially to the filling position during production.

Once the required initial conditions have been established, the operator can start the automatic production process using the Run command. The PLC and RobotStudio then operate together through the OPC UA communication.

7.2 Normal Production

During normal production, the input conveyor transports an automotive bracket towards the robot working area. When the workpiece reaches the pick position, the pick sensor stops the conveyor and the robot starts the handling sequence.

The robot picks the workpiece and transfers it to the simulated scanning station. After placing the workpiece in the scanner, the robot waits for three seconds while the simulated inspection is performed. The RAPID program then generates the classification result.

If the workpiece is classified as OK, the robot transfers it to the active tray and places it in the position indicated by the Tray_Hole value. If the workpiece is classified as NOK, the robot places it on the inclined NOK platform, from where the workpiece moves towards the NOK container.

After the corresponding handling operation has been completed, the robot returns to the Home position and the PLC updates the process information. The input conveyor can then provide the next workpiece.

For OK workpieces, the PLC also updates the number of pieces placed in the active tray. When six OK workpieces have been placed, the tray is considered full and the tray replacement sequence is initiated.

7.3 Automatic Tray Replacement

When the active tray reaches its capacity of six OK workpieces, the tray exit section of the conveyor is activated. The full tray moves out of the filling position and the Tray_Out signal is used to indicate that the tray has left the corresponding area.

The next tray in the buffer is then moved into the filling position. The tray sensors provide the position information required by the PLC to control the movement of the trays.

This sequence is repeated for the available trays in the buffer. The system therefore allows several trays to be filled sequentially without requiring the operator to manually replace the active tray after each filling operation.

When the fifth and final tray has left the cell, the system returns to the initial-condition state. The operator must then load a new set of empty trays before production can continue.

7.4 End-of-Cycle

The End-of-Cycle function allows the operator to request the end of production without stopping the process immediately. When the EoC command is activated, the corresponding process state changes to the End-of-Cycle condition.

The current workpiece handling sequence is allowed to continue, and the active tray can be completed. Once the current tray has left the cell, the production sequence stops.

To restart production, the operator must activate the Run command again. Since the tray-filling sequence is restarted, any remaining empty trays from the previous buffer configuration must be removed from the filling area before a new tray-filling sequence is initiated.

7.5 Quick Stop

The Quick Stop function provides a temporary interruption of the process. When the Stop command is activated, the robotic sequence is interrupted and remains stopped.

After the Run command is activated again, the sequence continues from the process point at which it was interrupted. This function was tested at different stages of the simulated process to verify that the sequence could be resumed correctly.

7.6 Emergency Stop and Rearm

The Emergency Stop function interrupts the normal operation of the simulated cell and activates the emergency state. The robot stops its current operation, while the state and tray lights indicate the emergency condition. The emergency alarm remains active while the emergency condition is maintained.

To resume operation, the emergency input must first be returned to its initial condition and the Rearm command must be activated. The operator must then activate Run to restart the process.

The simulated emergency sequence was tested at different points of the process. The system was able to resume the corresponding process sequence after the emergency condition had been cleared and the required rearming procedure completed.

The implemented Emergency Stop behaviour is limited to the simulation environment and does not represent a validated industrial safety function.

8. Engineering Challenges and Solutions

During the development of the cell, several technical challenges were identified and solved through an iterative development process. The main difficulties were related to the mechanical integration of the robot and tray, the simulation of the automatic tray buffer, the communication between the two software environments and the synchronisation of PLC and RobotStudio signals.

8.1 Robot and Tray Collisions

One of the first problems encountered during the development of the cell was the occurrence of collisions between the robot and the OK tray. The external walls of the initial tray design interfered with the robot trajectory during the placement of the workpieces.

The problem was identified by analysing the robot motion in RobotStudio. Since the tray geometry was intended for simulation rather than manufacturing, the design could be modified

directly in Siemens NX. Therefore, the OK tray was redesigned to provide sufficient clearance for the robot.

After modifying the tray geometry, the robot could reach the required tray positions without collisions. The problem was therefore solved through an iteration between the mechanical design and the robotic simulation.

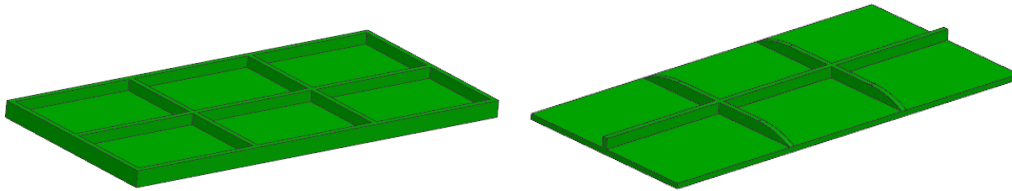


Figure 3. First Tray Design vs Final Tray Design

8.2 Automatic Tray Buffer and Workpiece Management

The most demanding challenge was the implementation of the complete automatic tray replacement process in the simulation. The difficulty was not only moving the trays through the conveyor, but also ensuring that the workpieces already placed in a tray moved together with the corresponding tray.

A conventional conveyor movement of the tray alone was therefore not sufficient. A mechanism was required to maintain the association between each tray and the workpieces placed in it.

The solution was implemented using a Smart Component Queue block in RobotStudio. An independent queue was created for the workpieces placed in the OK tray, so that only the workpieces belonging to the tray were added to this queue. When the tray moved through the exit section of the conveyor, the associated workpieces could therefore move together with it.

This solution allowed the complete tray replacement process to be simulated, including the movement of a full tray out of the filling position and the subsequent arrival of the next tray from the buffer.

8.3 OPC UA Communication Between Two Computers

Another significant challenge was establishing OPC UA communication between the Siemens and RobotStudio environments running on two different computers.

The initial configuration used Wi-Fi for both computers, but the communication did not operate correctly. The exact cause of this behaviour was not determined. Instead of further modifying the network configuration, an alternative connection arrangement was tested.

The computer running the OPC UA server was connected to the router using Ethernet, while the RobotStudio computer remained connected through Wi-Fi. With both computers connected to the same router, the OPC UA communication operated correctly.

The communication was subsequently verified using UaExpert by monitoring the PLC variables in real time and changing selected signal values to confirm that the changes were correctly applied. RobotStudio was then connected to the PLC to establish the complete bidirectional communication between both environments.

8.4 PLC and RobotStudio Synchronisation

During the integration of the PLC and RobotStudio, some process steps that had previously worked correctly did not always execute as expected. The problem was related to the timing of the exchanged signals.

In some cases, RobotStudio did not have enough time to read a signal before the PLC changed its value again. In other cases, the PLC required additional time to update a signal before RobotStudio read its new value.

The solution was to introduce additional waiting times in the RobotStudio program and timers in the PLC logic. These delays provided sufficient time for the corresponding system to detect and process the signal changes before the next operation was initiated.

This approach improved the synchronisation between both simulation environments and prevented process steps from being skipped due to signal timing.

8.5 Development Methodology

The challenges encountered during the project were addressed using an iterative development methodology. For each required function, the intended sequence was first considered conceptually and represented on paper. The corresponding solution was then implemented in the relevant software environment.

If the implemented solution did not behave as expected, the process was analysed to identify the reason for the failure. A modified solution was then implemented and tested again.

This methodology was particularly relevant for the OPC UA communication and the integration between TIA Portal and RobotStudio. Although basic knowledge of both RobotStudio and TIA Portal was available before starting the project, the communication between both environments through OPC UA and several of the functionalities used during the project were new.

9. Testing and Validation

9.1 Development Testing

Testing was performed continuously throughout the development of the cell. After implementing each relevant function, the corresponding part of the simulation was tested to verify its behaviour before continuing with the next development stage.

This iterative approach was used for the PLC logic, HMI, robot procedures, tray management and OPC UA communication. When a problem was identified, the implementation was modified and tested again until the expected behaviour was achieved.

The testing was mainly functional and was performed entirely within the simulation environment. Therefore, the results demonstrate the correct behaviour of the implemented simulation but do not constitute physical or industrial validation of the system.

9.2 Functional and Operating Condition Testing

Once the individual functions had been implemented, the main operating conditions of the cell were tested as part of the complete process. These included the initial tray-filling process, normal production, OK/NOK classification, all six OK positions in the tray, NOK discharge, automatic tray replacement and the five-tray buffer operation.

The End-of-Cycle, Quick Stop and Emergency Stop functions were also tested at different stages of the process. The corresponding behaviour of the robot, PLC and HMI was verified, including process resumption after Quick Stop and the rearm procedure following an Emergency Stop.

The OPC UA communication between the PLC and RobotStudio was tested during the integration of the two simulation environments. The exchanged signals were monitored using UaExpert and subsequently verified during the operation of the complete cell.

During the first integration tests, some timing and synchronisation problems were observed between the PLC and RobotStudio. Waiting times were therefore added to the RobotStudio sequence and PLC timers were introduced where required. The modified sequence was tested again until the process steps were correctly synchronised.

9.3 Complete System Test

After the individual functions and communication were verified, the complete simulated cell was executed from the initial tray-filling stage through normal production and automatic tray replacement.

The final test included workpiece feeding, robotic picking, simulated inspection, OK/NOK classification, tray filling, NOK discharge, tray replacement, HMI operation and the different stop conditions.

The complete sequence was successfully executed within the defined simulation scope. An executive video of approximately two minutes was recorded to document the main operating functions of the developed cell.

No formal production-performance KPIs, such as cycle time, throughput or efficiency, were calculated. The validation was therefore limited to functional verification of the implemented simulation and its interaction between the PLC, HMI and RobotStudio environments.

10. Limitations and Future Development

10.1 Project Limitations

The main limitation of the project is that the complete cell was developed and validated within a simulation environment. No physical robot, PLC, conveyor, sensor, gripper or other industrial hardware was implemented. Therefore, the results demonstrate the functional behaviour of the simulated system but do not provide physical validation of the mechanical components, robot operation or control system.

The inspection stage is also simulated. The classification result is generated in the RAPID program using a predefined probability, with approximately 90% of the workpieces classified as OK and 10% as NOK. This approach was sufficient to simulate the different process paths, but it does not represent the performance or accuracy of a real inspection system. No camera-based vision system was implemented.

Artificial intelligence and machine learning were not implemented in the final version of the project. Although these technologies could be used for more advanced inspection or process monitoring, their implementation was outside the defined scope of the project.

The mechanical design also has limitations related to its simulation-oriented purpose. The components were designed to verify the spatial integration of the cell and the robot movements,

but they were not developed as manufacturing-ready components. Detailed manufacturing drawings, tolerances, material specifications and structural calculations were not included.

The emergency handling implemented in the PLC and RobotStudio programs was developed to simulate the required operating behaviour. However, it does not constitute a validated industrial safety system. No physical safety devices or formal safety validation were implemented.

Finally, the testing was focused on functional verification of the implemented simulation. Formal production-performance analysis was not performed, and no industrial KPIs such as throughput, efficiency or availability were calculated.

10.2 Future Development

The developed simulation provides a basis for several possible future developments.

The first step would be the physical implementation of the cell. This would require the selection and integration of real industrial hardware, including the robot, PLC, conveyors, sensors, gripper and associated actuators. The mechanical components would also need to be redesigned and documented for manufacturing and physical assembly.

The simulated inspection system could be replaced by a real vision-based inspection system. A camera and suitable image-processing methods could be used to detect dimensional, positional or surface-related deviations in the workpiece. This would allow the OK/NOK classification to be based on actual inspection data rather than a simulated probability.

A further development would be the integration of Python-based data processing and artificial intelligence methods. These could be used to analyse inspection or process data and identify abnormal behaviour. For example, historical process data could be analysed to detect deviations that may indicate a problem with the gripper, workpiece positioning or other elements of the cell.

The safety functions would also need to be redesigned and validated for a physical industrial implementation. This would require appropriate safety-related hardware, control functions and validation procedures rather than the simulated emergency behaviour used in the current project.

Another development area would be the extension of the mechanical design towards a manufacturing-ready solution. This would include detailed drawings, tolerances, material selection and the engineering of the pneumatic and mechanical components required for physical operation.

Finally, the system could be integrated into a broader manufacturing environment. Production data could be collected and analysed to evaluate cycle times, throughput, equipment utilisation and other performance indicators. The OPC UA communication architecture could also be extended to connect the cell with higher-level manufacturing or monitoring systems.

These developments would transform the current simulation-based engineering concept into a physically validated and more fully integrated industrial manufacturing system.

11. Conclusions

This project achieved the development of a complete simulated automated pick-and-place cell integrating mechanical design, PLC programming, HMI development, robotic programming and industrial communication. The system combines Siemens TIA Portal and PLCSIM Advanced for

process control with ABB RobotStudio for robotic operation, while OPC UA enables bidirectional communication between both environments.

The developed cell successfully reproduces the main operating conditions of the proposed process, including workpiece feeding, robotic pick-and-place operations, simulated OK/NOK classification, tray filling and automatic tray replacement. End-of-Cycle, Quick Stop and Emergency Stop behaviours were also implemented and tested at different stages of the process. During development, several technical challenges related to robot-tray collisions, tray management, OPC UA communication and timing synchronisation were identified and resolved through an iterative engineering approach.

The project remains limited to a simulation environment. No physical hardware, real inspection system, AI/ML solution or validated industrial safety system was implemented. Therefore, further development would be required to transfer the concept to a physical manufacturing cell, including real inspection, industrial safety implementation, manufacturing-ready mechanical design and performance evaluation using production data.

Overall, the project demonstrates the feasibility of integrating robotic, PLC, HMI, mechanical and communication technologies into a single automated manufacturing concept, providing a basis for further development towards a physical and more advanced Industry 4.0 application.

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12.2 Online Tutorials and Learning Resources

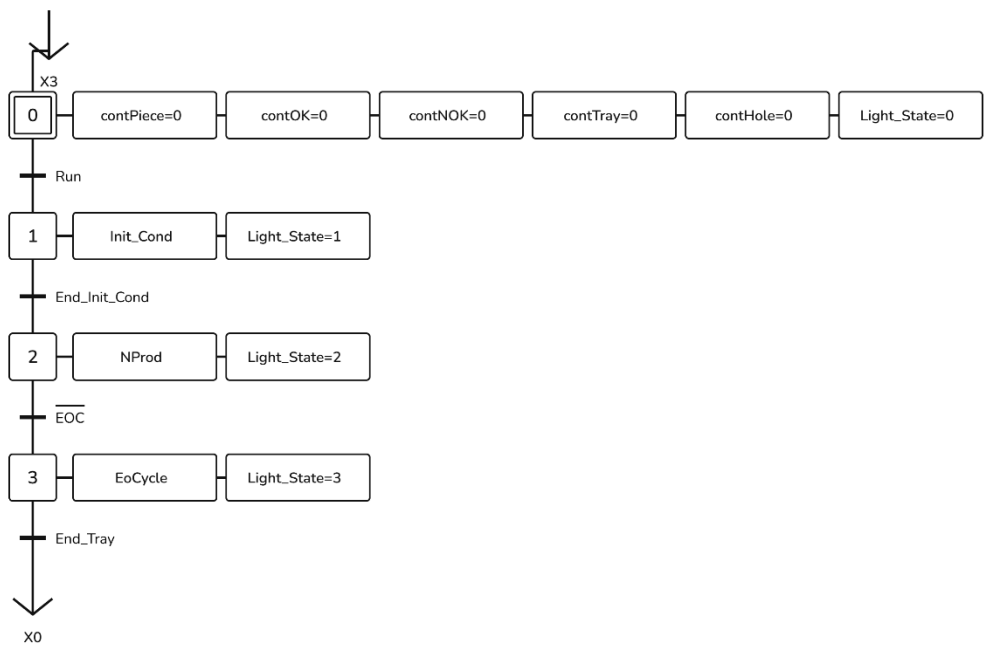
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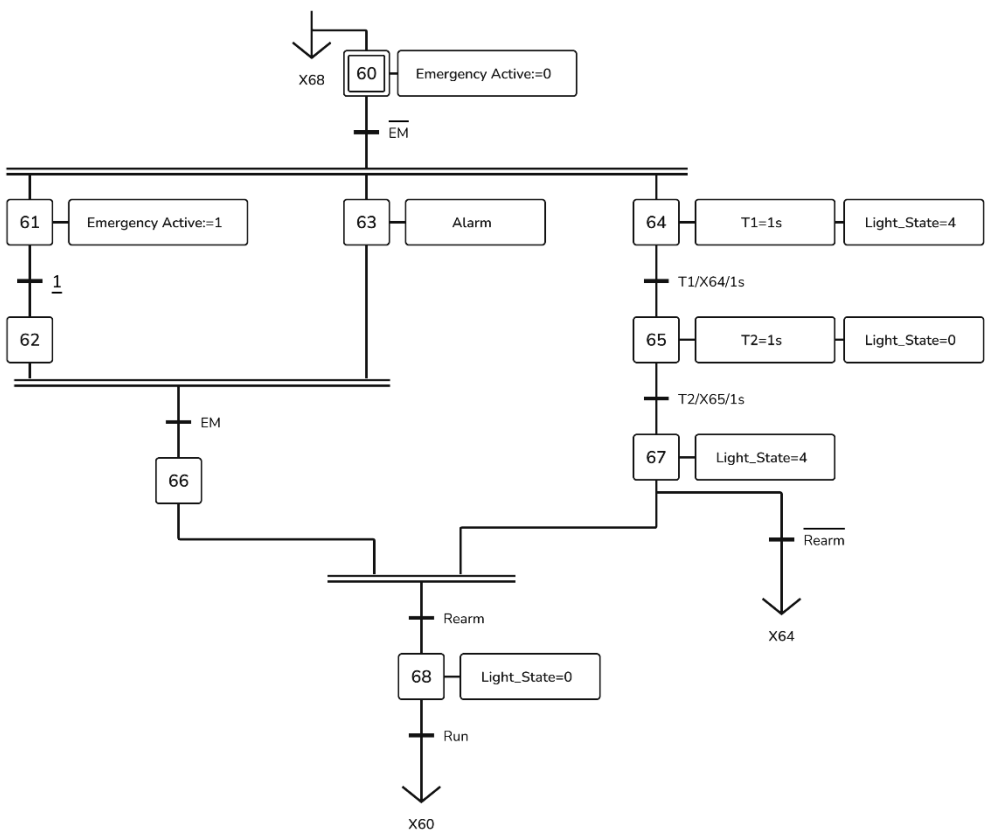
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Annex 1

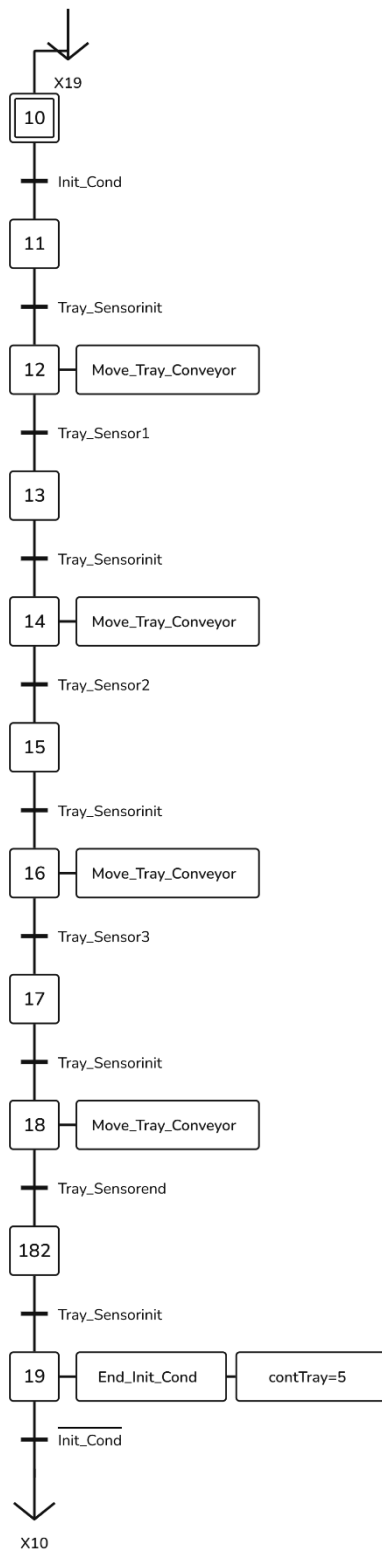
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